

Evaluation of Polypropylene/Polyolefin Elastomer Blends for Potential Recyclable HVDC Cable Insulation Applications

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ABSTRACT

This paper evaluates the microstructure and properties of polypropylene/polyolefin elastomer (PP/POE) blends for potential recyclable HVDC cable insulation applications. PP/POE blends with different POE content were prepared by melt mixing. The introduction of POE results in a slight decrease of the melting points but improves the flexibility of PP. Compared with PP, the volume resistivity of the blends shows a decrease at low loading of POE and starts to increase when the POE loading is higher than 15 wt%. After the introduction of POE, the DC breakdown strength is slightly decreased and the hetero space charge accumulation is enhanced. Although the electrical properties of the PP/POE blends are inferior to those of the pure PP, the enhanced flexibility, high volume resistivity, high breakdown strength as well as the excellent thermal properties make the PP/POE blends have the potential for HVDC cable application. The hetero space charge accumulation is still an issue, and further modification of the blends should be considered for suppressing the space charges.

Index Terms - Polypropylene, polyolefin elastomer, insulation materials, HVDC cable, space charge.

1 INTRODUCTION

MOST modern extruded high voltage cables use cross-linked polyethylene (XLPE) as the insulation materials. XLPE has excellent dielectric and thermo-mechanical properties and low dielectric loss, which is a relatively ideal material for cable insulation application. More importantly, XLPE has an apparently enhanced thermo-mechanical performance at high temperature because of the cross-linked macromolecular networks. However, the crosslinking process also results in thermosetting materials, making the XLPE be difficult to be recycled at the end of its lifetime. Nowadays, minimizing the impact of human activities on the environment is becoming much more important than ever. Therefore, it is urgent to develop new recyclable insulation materials for cables [1]. For insulation applications of extruded high voltage cables, the materials should not only have excellent electrical and thermal properties but also have enough mechanical strength and flexibility, particularly at low temperature. In addition, the space charge behavior should be considered in the case of HVDC insulation applications.

For DC cables, a number of studies of LDPE and its blends, HDPE and its blends and LDPE/HDPE blends have been reported [2, 3]. Although these materials can be recycled, most of them are not suitable for HVDC cable applications because of their poor electrical properties in the presence of DC voltage, particularly the unsatisfactory behavior of their resistivity vs. electric field and temperature. For an insulation candidate for HVDC cable application, the material should possess electrical resistivity in the working range of electric field and temperatures, better DC breakdown performance, less sensitive to electrical aging and space charge behavior [4-7].

Polypropylene which has high melting points and excellent electrical properties, is a good potential for cable insulation applications. Of great importance is that the polypropylene is a thermoplastic material, which is very easy to be recycled at the end of its lifetime. Nevertheless, the traditional polypropylene is not capable as cable insulation materials because of its stiffness and brittleness at room temperature, further modification should be studied for cable insulation applications.

From the point of view of increasing the operating electric field and temperature of extruded cables, some polypropylene based materials had been investigated. Yoshino et al investigated

the basic properties of syndiotactic polypropylene (sPP), which was synthesized with homogeneous metallocene catalyst [8]. It was found that both the thermal and electrical properties of sPP are superior to XLPE. When blended with VLDPE, sPP shows a sufficient brittleness temperature. Hosier et al investigated propylene-ethylene copolymers and their blends [1, 9, 10], the propylene-ethylene copolymers can improve the mechanical properties of PP but decrease electrical strength from their works. All of the propylene-ethylene copolymer materials exhibit one or more desirable properties but none of them possesses all the advantages. Furthermore, they found that blending is a good way to optimize the properties of different materials. According to the previous investigations, some critical issues should be resolved in the practical application of polypropylene in HVDC cable insulation. Previous investigations of electrical breakdown strength are usually carried out under an AC voltage. On the other hand, only few studies dealt with the space charge accumulation characteristics of polypropylene based materials.

Traditional rubbers like EPR or EPDM, which are called thermoplastic polyolefins (TPOs), are often used to improve the mechanical properties of PP, but with the development of catalyst in polyolefin synthesis, a new class of polymers was synthesized by using metallocene catalysts. This kind of synthetic polymers are called polyolefin elastomers (POEs), which shows a better performance on improving the mechanical properties of PP in previous study [11]. Most commercially available POEs are copolymers of either ethylene-butene, or ethylene-octene (EO), ethylbenzene (EB), styrene-ethylene-butadiene-styrene (SEBS). In this work, the ethylene-octene copolymer which is a kind of POE was added to improve the flexibility of polypropylene. The thermal, mechanical, thermo-mechanical and electrical properties of the PP/POE blends were investigated. The aim of this work is to explore the possibility of using PP/POE blends as recyclable HVDC cable insulation materials.

2 EXPERIMENTAL PROCEDURE

2.1 MATERIALS

The PP (F401) was purchased from Panjin Ethylene Co. Ltd., the isotacticity of which is 97 % and melt flow index 1.7-3.1 g/10 min, density 0.910 g/cm³, tensile yield strength 35 MPa. The POE (8150) was purchased from DuPont Dow, which is a kind of ethylene-octene copolymer with an octene comonomer content of 25 wt%, melt flow index 0.5 g/10 min, density 0.868 g/cm³, tensile strength 9.5 MPa, breaking elongation 810 %.

2.2 BLENDING AND SAMPLE PREPARATION

PP/POE blends with different POE content, as detailed in Table 1, were prepared by melt mixing. The required pellets of each material were weighed carefully using a digital balance and then were mixed in a HAAKE Rheometer RC90 at 190 °C and a rotor speed of 60 rpm for 6 minutes [12].

Film samples with different thickness were obtained using compression molding at 200 °C for 6 minutes under a pressure of about 15 MPa. Then the films were cooled to room temperature under the same pressure for 7 minutes. In order to

eliminate the thermal history of different films, all the films were annealed in the vacuum oven at 135 °C for 2 h and then cooled to room temperature in the vacuum. Then samples were stored in a vacuum drier to exclude the influence of moisture. For the convenience of description, we use the designations in Table 1 to represent the blends with different POE content.

Table 1. Blends prepared in this study.

Designation	PP Content (wt%)	POE Content (wt%)
PP0	100	0
PP5	95	5
PP10	90	10
PP15	85	15
PP20	80	20
PP25	75	25
PP30	70	30

2.3 CHARACTERIZATION

For observing the dispersion of POE in PP matrix and the microstructure of the blends, the cross sections of the blends were immersed in n-heptane at 50 °C for 20 minutes to remove POE and then washed with alcohol. After that the etched cross sections were sputter-coated with gold and then imaged by a SEM (JSM-6460LV). A differential scanning calorimeter (DSC) (NETZSCH 200F3) was used to study the thermal properties and crystallinity of the blends under nitrogen atmosphere at a heating/cooling rate of 10 °C/min between 20 and 200 °C. Tensile tests were undertaken according to ASTM D 638-2003 [13] using a tensile tester (CMT4304) with a fixed speed of 50 mm/min at 25 °C. Five specimens for each blend were used for repeat measurements. Dumbbell shaped specimens with a cross section of 4 mm × 1 mm and gauge length of 20 mm were prepared for the measurements. A dynamic mechanical thermal analyzer (DMTA) (DMA Q800) was used to study the thermo-mechanical properties of the blends. Samples (70 mm × 10 mm × 1 mm) were cut down from films and measurements were done at 1Hz between -60 and 150 °C at a heating rate of 3 °C/min in tension mode.

DC breakdown tests were performed on a DC dielectric strength tester (DH, Shanghai Lanpotronics Co., China) with an increasing DC voltage of 1 kV/s. The specimens (of thickness around 130 μm) were placed between two 30-mm-diameter copper ball electrodes and immersed in the pure silicone oil [14]. The results from 20 tests for each blend were treated by using two-parameter Weibull statistical distribution method [15]. The 90 % confidence intervals for Weibull parameters were calculated according to IEEE Std 930-2004 [16]. DC leakage current measurements were taken on film samples (of thickness around 130 μm) using three-electrode method [17] under a DC electric field of 5 kV/mm. The leakage current was recorded for 10 minutes using a Keithley 6517A and five times for each blend.

The space charge measurements were performed with a pulse electro-acoustic (PEA) system [18] under 30 kV/mm DC electric field for 1 h. Before measurements, all film samples (of thickness around 260 μm) were heat-treated at 135 °C in the vacuum oven for 2 h while electrically short-circuited to eliminate the charges generated during processing. A pulse with a voltage of 800 V and a width of 5 ns was applied to the film samples.

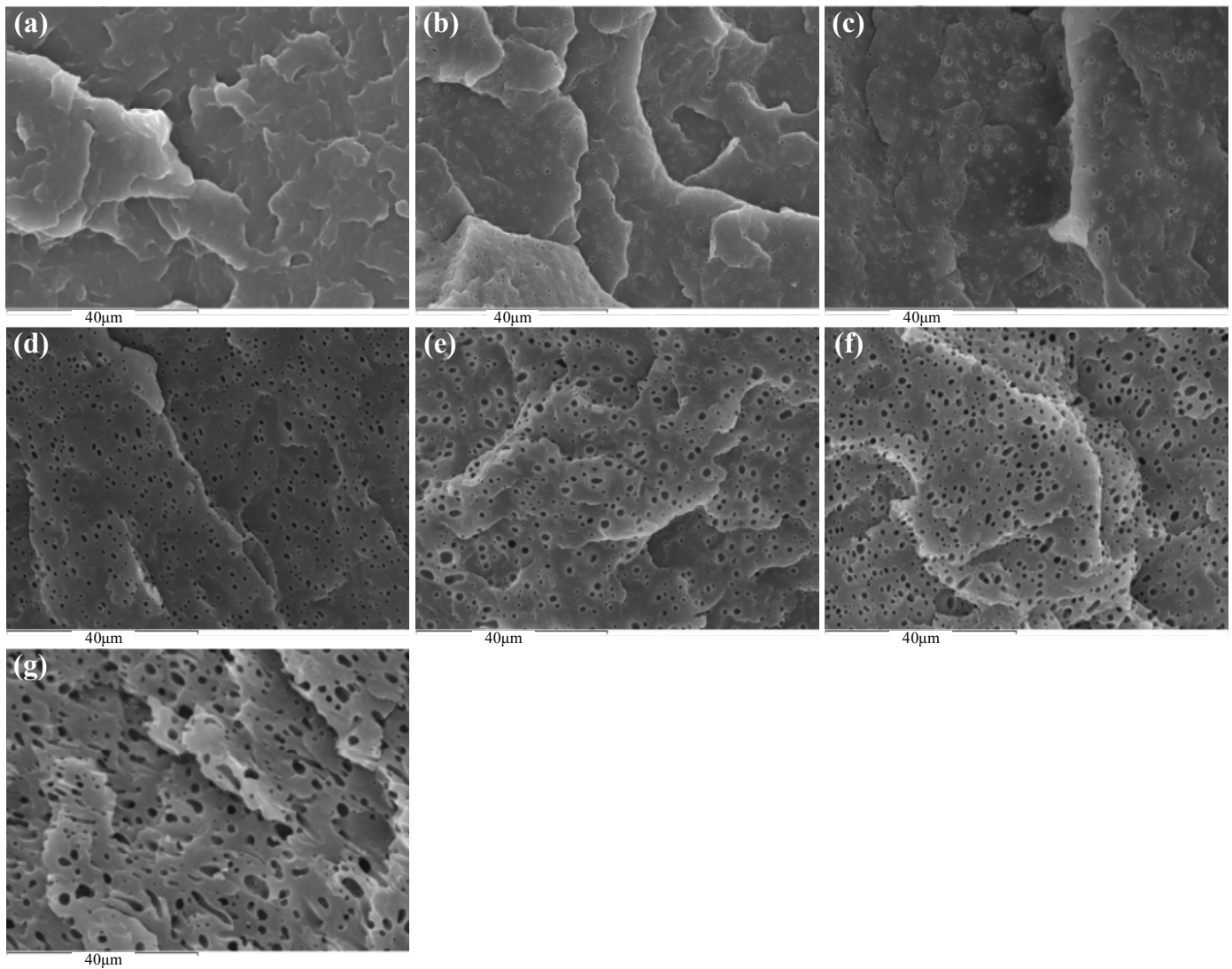


Figure 1. SEM images showing the dispersion of POE and the microstructure of the PP/POE blends. (a) PP0; (b) PP5; (c) PP10; (d) PP15; (e) PP20; (f) PP25; (g) PP30.

3 RESULTS AND DISCUSSION

3.1 DISPERSION AND MORPHOLOGY

The dispersion of POE in PP was studied by SEM. Figure 1 shows the etched cross sections of the pure PP and the PP/POE blends. The round holes in these figures should be attributed to the removal of POE by etching. As a result, the dispersion of these holes can represent the dispersion of POE in PP matrix. From this point of view, POE in PP matrix is nearly spherical with a diameter around 1 μm and forms a typical sea-island structure. POE is uniformly dispersed in PP, indicating that POE and PP have a certain degree of compatibility but not perfectly compatible. Figures 1f and 1g show that some of the POE particles are interconnected with each other when POE content is beyond 25 wt%, indicating the POE content is close to the critical point of brittle-ductile transition for the blends.

3.2 THERMAL ANALYSIS

The melting and crystallization curves of the blends are exhibited in Figure 2. Table 2 summarizes the melting temperature T_m , crystallization temperature T_c , melting enthalpies ΔH_m , crystallization enthalpies ΔH_c and crystallinity X_c of the blends. The melting temperatures of the blends exhibit a slight decrease when the POE content increases, indicating that the blends have good thermal properties. The decrease of melting temperature of PP with the increase of POE content suggests that PP and POE have a certain degree of compatibility according to the theory of polymer compatibility. This is consistent with the results of SEM observation. While the crystallization temperatures of the blends decrease with the increase of POE content, indicating weak nucleating ability of PP in the blends. Overall, the introduction of POE results in a decrease of crystallinity of the blends. This should be attributed to the interactions between the POE and the PP macromolecules, which makes the crystallization become more difficult.

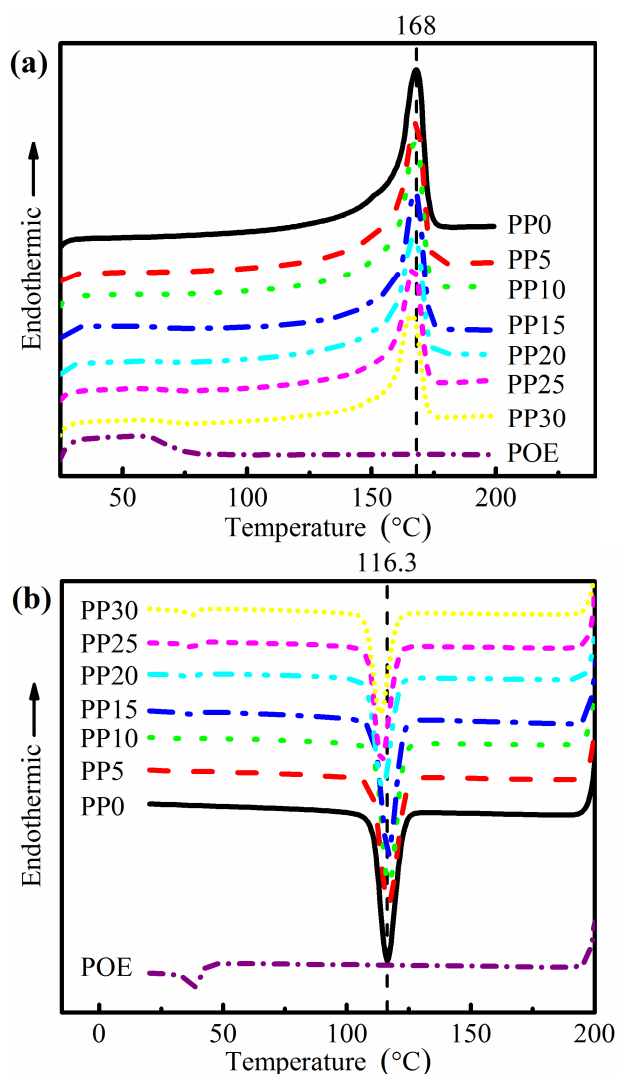


Figure 2. (a) DSC melting curves and (b) crystallization curves of the PP/POE blends.

Table 2. Summary of DSC data of the PP/POE blends

	PP0	PP5	PP10	PP15	PP20	PP25	PP30	POE
T_m (°C)	168.0	167.8	167.6	167.4	167.1	166.8	166.0	56.2
T_c (°C)	116.3	116.4	116.6	116.3	114.4	114.0	113.5	38.8
ΔH_m (J/g)	114.5	104.3	103.5	98.6	91.4	78.7	74.8	25.1
ΔH_c (J/g)	116.1	107.9	104.3	99.0	86.6	83.5	78.3	18.1
X_c (%)	54.80	49.89	49.52	47.17	43.75	37.66	35.80	8.57

3.3 MECHANICAL BEHAVIORS

Mechanical property is another important one to be considered in choosing a cable insulation material. Figure 3 shows the typical tensile stress-strain curves of the PP/POE blends. Furthermore, the breaking elongation, tensile yield strength and elastic modulus of the blends are listed in Table 3. The yield points shift to higher strains as the POE content increases. The tensile behaviors of the blends are much more similar to that of elastomers with the increase of POE content. The tensile yield strength decreases and the breaking elongation increases as the POE content increases. The decrease of tensile yield strength should be mainly attributed to the lower tensile yield strength of the POE. On the other hand, the decrease of crystallinity also results in a decrease of tensile yield strength, which is consistent with the results from

the DSC tests. POE has a great enhancement on the breaking elongation of the PP/POE blends, which should be mainly attributed to the rubber elastic properties of the POE. The higher breaking elongation indicates that the blends have sufficiently high mechanical flexibility for cable insulation applications. From Figure 3, the breaking elongation increases rapidly when the POE content is beyond 25 wt%, indicating that there is a brittle-ductile transition behavior for the blends. This also can be verified from the microstructure observation of the blends as shown in Figure 1.

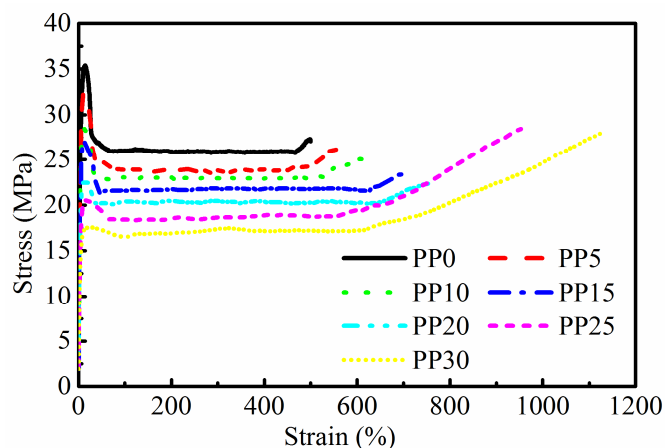


Figure 3. Tensile stress-strain curves of the PP/POE blends at 25 °C.

Table 3. Summary of tensile test data of the PP/POE blends.

Designation	Breaking Elongation (%)	Tensile Yield Strength (MPa)	Elastic Modulus (MPa)
PP0	501±36	35.0±1.7	586±37
PP5	555±41	32.3±1.6	595±28
PP10	620±32	28.3±1.5	523±36
PP15	702±31	25.1±1.8	461±39
PP20	761±37	22.9±2.0	422±35
PP25	953±48	20.9±1.3	340±26
PP30	1128±44	17.6±1.3	287±19

3.4 THERMAL MECHANICAL ANALYSIS

Figure 4 shows the storage modulus and mechanical loss tangent $\tan\delta$ as a function of temperature. DMTA results of the blends are consistent with those derived from the tensile tests. Namely the introduction of POE significantly enhances the flexibility of the PP/POE blends. As shown in Figure 4, the inclusion of POE serves to lower the modulus of the blends. The reason for this phenomenon is as follows. First, the result of DSC shows that the crystallinity of the blends decreases with POE content increases, which can reduce the storage modulus. Second, POE is a kind of elastomer with low storage modulus and as a result, the storage modulus of the blends decreases as the POE content increases.

It can be seen from Figure 4b that the introduction of POE leads to a new transition peak (Peak 1) at around -50 °C, which should be attributed to the glass transition of POE. On the other hand, the transition peak (Peak 2) of PP at about 10 °C shifts to higher temperatures with the increase of POE content. The higher transition temperatures of Peak 2 further indicate that there exist interactions between PP and POE. It is believed that the lower glass transition temperature of POE is the important factor resulting in the highly enhanced flexibility and low modulus of the PP/POE blends.

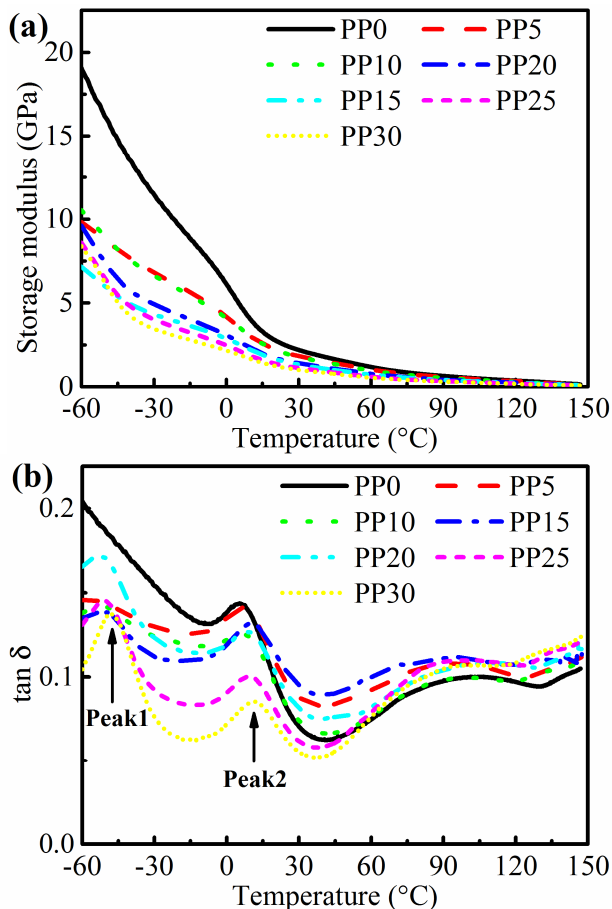


Figure 4. (a) storage modulus and (b) loss factor $\tan \delta$ of the PP/POE blends.

3.5 DC BREAKDOWN STRENGTH

Figure 5 shows the cumulative probability of DC electrical breakdown strength of the blends, which shows a slight decrease when the POE content increases. For example, the breakdown strength only shows a decrease of 10 % when the POE content is 30 wt%, indicating a high enough breakdown strength of the blends compared to that of the typical XLPE of 300 kV/mm [19]. Table 4 lists the Weibull parameters of the blends. The shape parameters of all blends are larger than 10, suggesting a low dispersion of the test data.

It has been reported that the electrical breakdown strength of polyolefin is affected by the thickness of lamellae and higher lamellae thickness usually results in higher breakdown strength [20]. Tashiro et al considered that the thickness of lamellae depends on the crystallization temperature and higher crystallization temperatures lead to thicker lamellae [21]. In this study, the DSC result shows that the crystallization temperature of the blends decreases with the increase of POE content, indicating that the lamellae thickness decreases and thus the DC breakdown strength decreases. Second, the introduction of POE may cause an increase of free volume, which increases the mean free path of electrons and thus causing a decrease of the electrical breakdown strength. Third, as shown in Figure 1, POE and PP are not perfectly compatible in the blends, causing numerous interface and defects which can trap

charges, leading to significant distortion of electric field and thus causing lower breakdown strength. This can be further verified by the following analysis of space charge accumulation of the blends.

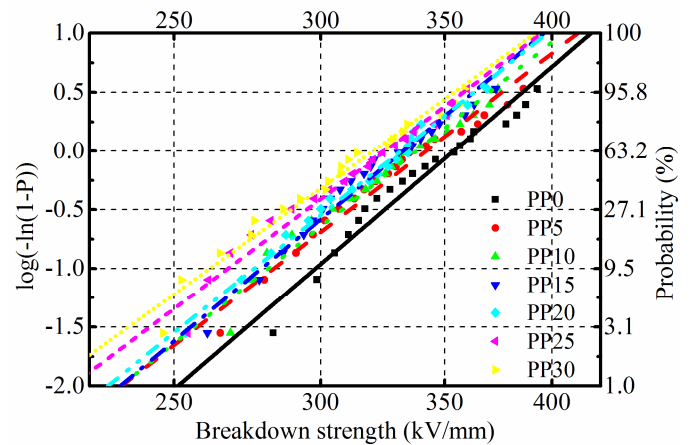


Figure 5. The Weibull plots for DC breakdown strength of the PP/POE blends at 25 °C.

Table 4. Weibull parameters for the PP/POE blends.

Designation	Scale Parameter E_0 (kV/mm)	90 % confidence interval for E_0	Shape Parameter β	90 % confidence interval for β
PP0	354.56	(343.23, 364.81)	12.84	(9.66, 18.57)
PP5	342.37	(330.50, 353.14)	11.82	(8.89, 17.09)
PP10	337.89	(326.48, 348.23)	12.14	(9.13, 17.55)
PP15	332.66	(321.92, 342.38)	12.71	(9.56, 18.38)
PP20	331.50	(320.63, 341.34)	12.51	(9.41, 18.09)
PP25	325.44	(313.78, 336.03)	11.43	(8.60, 16.53)
PP30	320.54	(308.65, 331.35)	11.03	(8.29, 15.95)

3.6 VOLUME RESISTIVITY

The leakage currents of the blends under 5 kV/mm DC electric field are shown in Figure 6a. It can be seen that the polarization currents have been completely decayed at 10 minutes and the leakage currents are all conduction currents. According to the leakage current at 10 minutes, the volume resistivity is calculated and shown in Figure 6b. It can be found that with the increase of POE content, the volume resistivity of the blends decreases at first and then increases. Comparing the volume resistivity of pure PP, pure POE and the blends, the resistivity of the blends is not simply the summation of the individual components because the volume resistivity of the blends is higher than both raw materials in some cases.

The results of DSC show that the crystallinity of the blends decreases with the increase of POE content. Therefore, the free volume between the crystalline regions increases, which can result in the increase of the mean free path of charge carriers and thus increase the mobility of the charge carriers. As a result, the volume resistivity of the blends decreases. On the other hand, the numerous defects at the interface of PP and POE can trap charge carriers and thereby the resistivity of the blends increases. From the analysis above, the resistivity of the PP/POE blends is affected by multiple factors and shows a decrease at low loading of POE and starts to increase when the POE loading is beyond 15 wt%.

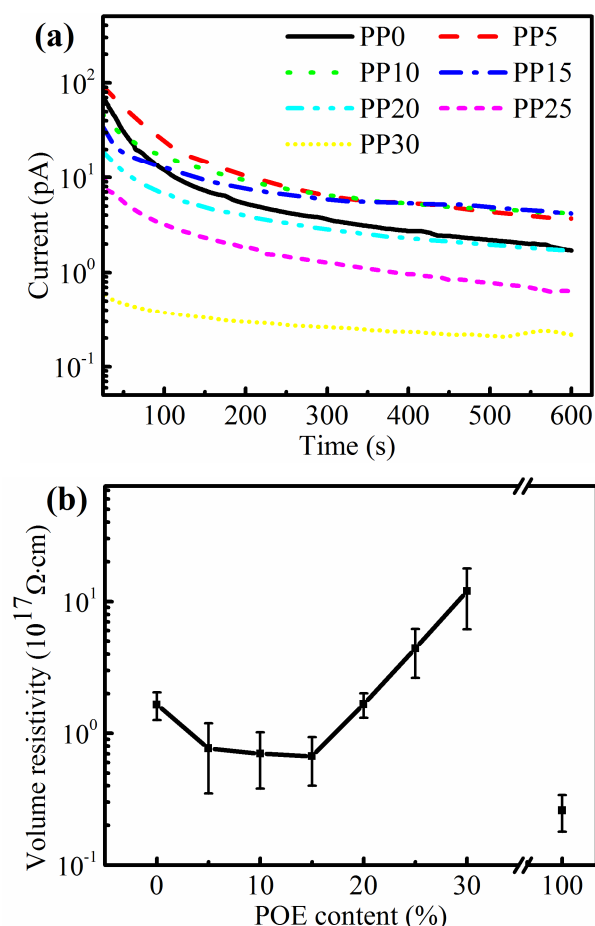


Figure 6. (a) Leakage current and (b) DC volume resistivity of the PP/POE blends at 25 °C

3.7 SPACE CHARGE DISTRIBUTION

For HVDC cables, the space charge accumulation in the insulation is one of the important factors causing the cable failure. The accumulation of space charge inside the insulation causes local distortion of the electric field and thus partial discharge and breakdown of the insulation [22]. Therefore, the space charge accumulation in HVDC cable insulation should be as low as possible.

Figure 7 shows the space charge distribution of the blend films within 1 h at 25 °C. Figure 7a describes the development of space charge accumulation in pure PP, a small amount of hetero space charge accumulates in the vicinity of cathode within 1 h. But the change of the space charge accumulation is not obvious over time, indicating that the charge accumulates quickly and reaches its steady state in a short period of time. Since the manufacturing process would inevitably introduce impurities and the raw materials also contain some impurities (i.e., catalysts), there exist some impurities and weakly bound ions in the blend films, which can polarize or ionize under DC electric field. Meanwhile, charges would be injected from the electrode into the material. The charges originating from above sources can be trapped and then space charge accumulation takes place [23]. Since the positive ion is heavier than the electron, the mobility of the positive ion is much lower than that of the electron with the same electricity

under the same applied electric field. Thus the positive space charge accumulation forms near the cathode in the pure PP.

Comparing the space charge accumulation behaviors of different blends, we can find that the space charge accumulation significantly increases with the increase of POE content. The amount of space charge and charge injection depth increases with time. This can be explained from the microstructure and crystal morphology of the blends. With the increase of POE content, a large amount of trapping sites can be introduced at the interface between POE particles and PP matrix. Also the semicrystalline nature of PP leads to the presence of trapping sites at the interface between the crystalline and amorphous phases [24]. As a result, the injected charges are captured by the trapping sites so that there are large amount of hetero space charge that accumulates near the electrode. Meanwhile, an independent electric field is formed by the hetero charges and the effective electric field inside the samples is changed, which results in the enhancement of the electric field near the electrode. Thus charge injection from the electrode is easier and charge injection depth increases. It should be noted that there is only a few hetero space charge accumulates in pure POE just like the space charge behavior of pure PP, so it can be concluded that the space charge in the blends is basically not caused by the impurities in pure POE and PP. It can be seen that the space charge accumulation is severe in the PP/POE blends prepared in this study, which can result in severe distortion of the electric field inside the material and insulation failure. These results mean that, for using as HVDC insulation, the PP/POE blends should be further modified to suppress the space charge accumulation.

3.8 COMPARISON BETWEEN PP/POE BLENDS AND XLPE

In order to better illustrate the properties of PP/POE blends, a comparison of the various performance considered in this investigation between PP/POE blends and typical XLPE is given in Table 5.

Table 5. Comparison between PP25 and XLPE.

Properties	PP25	XLPE
Tolerable temperature	≥ 120 °C	About 105 °C
	melting temperature	softening temperature [25,26]
Breaking elongation	953 %	600 % [27]
Tensile yield strength	20.9 MPa	22.0 MPa [27]
DC breakdown strength	325 kV/mm	300 kV/mm [19]
DC volume resistivity	$4.42 \times 10^{17} \Omega \cdot \text{cm}$ (5 kV/mm)	$0.92 \times 10^{17} \Omega \cdot \text{cm}$ (10 kV/mm) [23]
Space charge	Serious hetero space charge accumulation and injection near cathode (30 kV/mm)	Small amount of hetero space charge near anode and slight charge injection near cathode (-50 kV/mm) [23]

The tolerable temperature of PP25 is higher than XLPE indicating a better thermal performance. The mechanical properties of PP25 is effectively increased by the introduction of POE. The dielectric properties of PP25 is comparable or even higher than XLPE (i.e., increased volume resistivity and high breakdown strength). The space charge accumulation in PP25 is more serious than typical XLPE.

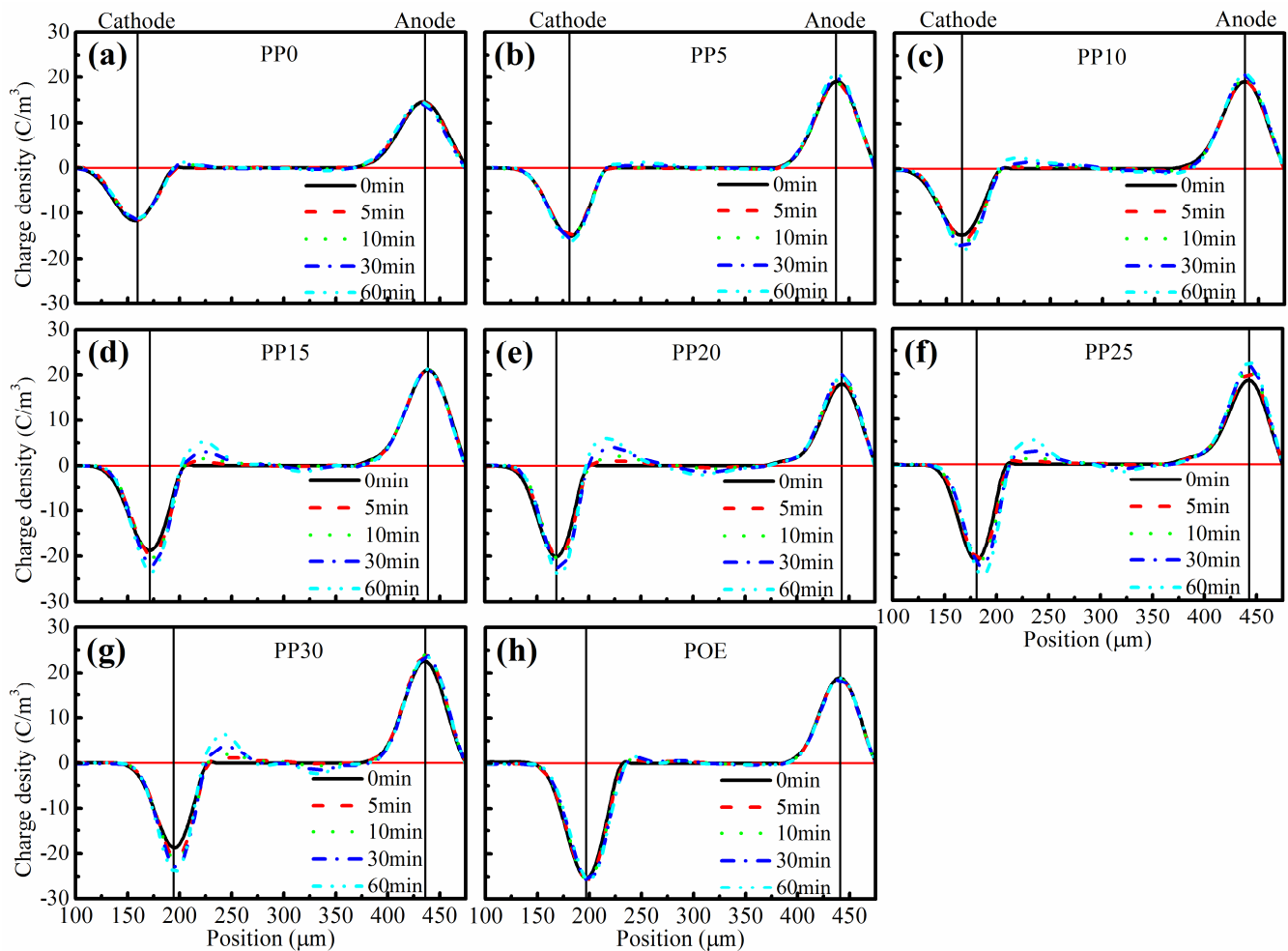


Figure 7. Space charge distribution during polarization of samples subjected to a DC electric field of 30 kV/mm at 25 °C for 1 h. (a) PP0; (b) PP5; (c) PP10; (d) PP15; (e) PP20; (f) PP25; (g) PP30; (h) POE.

4 CONCLUSIONS

In this study, the possibility of PP/POE blends used in recyclable HVDC cable insulation system was investigated. Blend materials with different POE content were prepared by melt mixing. The microstructure, thermal, mechanical, thermo-mechanical, DC electrical and space charge accumulation properties of the blends were studied. The effect of POE content on the properties of the PP/POE blends can be summarized as below:

(a) The PP/POE blends show a typical sea-island structure and the dispersion of POE is uniform. Some POE particles interconnects and the blends start to show a brittle-ductile transition when the POE content is beyond 25 wt%.

(b) The melting points of the blends slightly decrease after the inclusion of POE but the thermal properties are still good enough. The crystallinity of the blends decreases to 36 % of the blends with 30 wt% POE from 55 % of the pure PP.

(c) The tensile tests show that the POE can effectively improve the flexibility of PP to meet the requirements of HVDC cables.

(d) The introduction of POE can result in the decrease of storage modulus of the blends, whereas the glass transition temperatures of PP are increased because of the interaction between PP and POE. The results of DMTA show that the blends

not only have adequate mechanical integrity at high temperature but also possess flexibility at low temperature.

(e) The DC dielectric strength of the blends shows a slight decrease as the POE content increases and only decreases by 10 % when POE content is 30 wt%.

(f) With the increase of POE content, the volume resistivity of the blends decreases at first and then increases, but is still comparable to XLPE.

(g) A small amount of hetero space charge accumulates near the cathode in pure PP. The space charge accumulation significantly increases with the increase of POE content due to the increase of trapping sites introduced at the interface between POE and PP matrix. This can severely affect the electrical properties and long-time performance of the material.

Taking the aforementioned results into account, it is concluded that the introduction of POE can result in enhanced mechanical flexibility, good enough thermal properties (i.e., high melting temperatures) and enhanced electrical properties (i.e., increased volume resistivity and high breakdown strength). Therefore, PP/POE blends are promising candidates for recyclable HVDC cable insulation applications. But the hetero space charge accumulation in the blends is still a limit of the practical application of PP/POE blends. Further work should be done to suppress the space charge accumulation.

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